Arham Lodha
Math 215A

Homework 5
Due: October 28, 2026

Rings are understood to be commutative, unless stated otherwise.

Problem 1

Give an example of a ring A and a multiplicatively closed subset $S \subset A$ such that the relation "$(a, s) \sim (b, t)$ if $at - bs = 0 \in A$" on $A \times S$ is not an equivalence relation. [This explains the more complicated definition of $A[S^{-1}]$, in general. Make sure you know that definition.]

Solution. Consider $A = \mathbb{Z}_6$ and $S = \{1, 3\}$. Consider $(0, 3), (0, 1), (2, 3)$.

$$(0) * 1 - 3 * 0 = 0 \Rightarrow (0, 3) \sim (0, 1)$$
$$0 * 3 - 3 * 2 = 6 = 0 \Rightarrow (0, 3) \sim (2, 3)$$
$$0 * 3 - 1 * 2 = -2 = 4 \neq 0 \Rightarrow (0, 1) \text{ is not equivalent to } (2, 3).$$

Thus $\sim$ is not transitive. Hence it cannot be a equivalence relation.

Problem 2

Let A be a ring, $f \in A$, and M an A-module. Show that the localization $M[1/f]$ is isomorphic to the direct limit

$$L = \lim_{\rightarrow} \Big(M \xrightarrow{f} M \xrightarrow{f} M \xrightarrow{f} \cdots \Big).$$

Proof:

$$\lim_{\rightarrow} \Big(M \xrightarrow{f} M \xrightarrow{f} .... \Big) = \bigsqcup_{k=1}^{\infty} M / \sim .$$

where for all $j \geq i$ and $x \in M_i$ and $y \in M_j$, $x \sim y \Leftrightarrow$. Note that

$$M\left[\frac{1}{f}\right] \cong M \otimes A\left[\frac{1}{f}\right].$$

$$T : L \to M\left[\frac{1}{f}\right]$$
$$x_i \to \frac{x_i}{f^i}$$

Well defined: Suppose $x_i \sim y_j$. Without loss of generality $j \geq i$, thus $f^{j-i}x = y$. Thus

$$T(y_j) = \frac{y}{f^j}$$
$$= \frac{f^{j-i}x}{f^j}.$$

Note that

$$f^{j-i}x * f^i - f^j * x = f^j x - f^j x = 0.$$

Thus

$$T(y_j) = \frac{f^{j-i}x}{f^j} = \frac{x}{f^i} = T(x_i).$$

Thus the map is well defined.

Injective: Suppose $T(x) = 0$. Then

$$0 = \frac{x}{f} \Rightarrow \frac{0}{f^k} = \frac{x}{f}.$$

for some $k \in \mathbb{N}_0$. $f^k x - f * 0 = 0 \Rightarrow f^k x = 0 \Rightarrow x = 0 \in L$.

Surjective: $\forall r \in M\left[\frac{1}{f}\right]$,

$$r = \frac{m}{f^k}$$

for $m \in M$ and $k \in \mathbb{N}_0$. $T(m_k) = \frac{m}{f^k}$.

Thus the map is a isomorphism as a $A\left[\frac{1}{f}\right]$ module. □

Problem 3

Let R be a ring, $f \in R$. Show that the localization $R[1/f]$ is isomorphic to $R[x]/(fx - 1)$ (as an R-algebra).

Problem 4

Let R be a ring, $S \subset R$ a multiplicatively closed subset, M an R-module. Suppose that for every $s \in S$, multiplication by s is a bijection from M to itself. Show that the natural map $M \to M[S^{-1}]$ is an isomorphism. As a sample application, show that the localization of the $\mathbb{Z}$-module $\mathbb{Z}/8$ at the prime ideal $(2) \subset \mathbb{Z}$ is simply $\mathbb{Z}/8$.

Problem 5

A *finite algebra* over a ring A means an A-algebra which is finitely generated as an A-module. Show that a finite flat $\mathbb{Z}$-algebra is free as a $\mathbb{Z}$-module, and give an example of a flat $\mathbb{Z}$-algebra which is not free as a $\mathbb{Z}$-module. Is a flat $\mathbb{Z}$-algebra of finite type always free as a $\mathbb{Z}$-module?